

418-hw5

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Problem 1

```
sig = matrix(c(3,0,0,0,9,0,0,0,6),3,3,byrow=TRUE)
pca = princomp(sig)
pca$loadings
```

```
##
## Loadings:
##      Comp.1 Comp.2 Comp.3
## [1,]  0.101  0.505  0.857
## [2,] -0.909 -0.303  0.286
## [3,]  0.404 -0.808  0.429
##
##              Comp.1 Comp.2 Comp.3
## SS loadings      1.000  1.000  1.000
## Proportion Var   0.333  0.333  0.333
## Cumulative Var   0.333  0.667  1.000
```

```
# a
#  $Y_1 = 0.101x_1 - 0.909x_2 + 0.404x_3$ 
#  $Y_2 = 0.505x_1 - 0.303x_2 - 0.808x_3$ 
#  $Y_3 = 0.857x_1 + 0.286x_2 + 0.429x_3$ 

# b
# By definition, the covariance (and thus correlation) between any two of the
# principal components is 0. That said, perhaps the most important observation
# is that all three components involve all of the original variables. None of
# the three variables were found to be redundant of the others, or in other
# words there is no multicollinearity in the data based on the sigma matrix.
# This could have been anticipated though, as the covariance between any pair
# of  $x_i$ 's was already known to be 0. If there was any high correlation between
# any of the variables, one might anticipate data reduction (due to collinearity)
# to be more necessary. Generally, there is no multicollinearity in the dataset
# so principal component analysis is not necessarily needed (and is in fact
# somewhat redundant as it only created more variables without demonstrating the
# non-necessity of others).

# c
summary(pca)
```

```
## Importance of components:
##               Comp.1   Comp.2   Comp.3
## Standard deviation   4.582576 2.645751 8.429370e-08
## Proportion of Variance 0.750000 0.250000 2.537653e-16
## Cumulative Proportion 0.750000 1.000000 1.000000e+00
```

Based on the summary output, it can be seen that the first component explains 75% of the total variance in the population, the second explains 25% of the total variance, while the third explains an extremely negligible amount. If one were to choose which principal components to include, they would likely use only the first two components as they essentially explain 100% of the variability in the population.

Problem 2a

```
radio = read.csv('C:/Users/jacma/OneDrive/Documents/418hw5.csv')
fix = apply(radio[-1,],2, as.numeric) # remove first row (strings), as numeric
# a
S = cov(fix)
R = cor(fix)
cat('Dispersion/Covariance Matrix: \n')
```

```
## Dispersion/Covariance Matrix:
```

S

```
##           x1           x2           x3           x4           x5           x6
## x1 4.6547509  0.93134537 0.58969909 0.27691531  1.074885659  0.158150852
## x2 0.9313454  0.61282116 0.11093341 0.11846905  0.388886434 -0.024851988
## x3 0.5896991  0.11093341 0.57142886 0.08700496  0.347989910  0.110131391
## x4 0.2769153  0.11846905 0.08700496 0.11040907  0.217405649  0.021814433
## x5 1.0748857  0.38888643 0.34798991 0.21740565  0.862172372 -0.008817694
## x6 0.1581509 -0.02485199 0.11013139 0.02181443 -0.008817694  0.861455923
```

```
cat('Correlation Matrix: \n')
```

```
## Correlation Matrix:
```

R

```
##           x1           x2           x3           x4           x5           x6
## x1 1.00000000  0.55143669 0.3615773 0.38627479  0.53655840  0.07897812
## x2 0.55143669  1.00000000 0.1874625 0.45544470  0.53500626 -0.03420407
## x3 0.36157729  0.18746250 1.00000000 0.34638617  0.49577944  0.15696886
## x4 0.38627479  0.45544470 0.3463862 1.00000000  0.70464665  0.07073348
## x5 0.53655840  0.53500626 0.4957794 0.70464665  1.00000000 -0.01023155
## x6 0.07897812 -0.03420407 0.1569689 0.07073348 -0.01023155  1.00000000
```

```
# From the output it can be seen that some of the variables are more highly
# correlated than others. Namely, the correlation between x4 (eat) and x5
# (appetite) is higher than 0.7, suggesting there may be multicollinearity
# between them. For that matter, x5 seems to be fairly correlated with each
# variable aside from x6 (skin reaction). Additionally, x6 has its strongest
# correlation with x3 (sleep), but very negligible relationships with all of the
# other variables (correlation is close to 0). Based on this reasons, one might
# anticipate the principal component analysis to omit x5 but keep x6. This is
# because the former is somewhat redundant of the other variables, but the
# latter is barely redundant at all. For clarity, almost all variables seem to
# be somewhat related, the lowest correlation aside from x2 with x3 is 0.34
# between x3 and x4. Thus, there is reason to believe principal component
# analysis can be used to reduce the dimensionality of the data.
```

Problem 2b

```
# b
# The following principal component analysis will be based off the covariance
# matrix as there does not appear to be any issues with the scales of the
# variables. Additionally, using the correlation matrix can make determining the
# cutoff point regarding which components to use more difficult. Standardizing
# each variable typically means that more variables will be involved in the
# principal components, potentially defeating the purpose of PCA.
eigendecom = eigen(S)
evals = eigendecom$values
cat('Eigenvalues: \n',evals,'\n')
```

```
## Eigenvalues:
## 5.283381 0.8873149 0.7675244 0.434911 0.2495257 0.05038164
```

```
evecs = eigendecom$vectors
cat('Eigenvectors: \n')
```

```
## Eigenvectors:
```

```
evecs
```

```
##          [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
## [1,] 0.92848514 -0.01984119 0.34897158 -0.11153020 -0.05642494 0.01085087
## [2,] 0.21161527 0.16899493 -0.22680309 0.71009823 0.60716267 -0.04831192
## [3,] 0.14237865 -0.15094403 -0.50211919 -0.63510070 0.54883906 0.01559889
## [4,] 0.06797085 0.01498160 -0.20314436 0.08586371 -0.12741820 0.96451019
## [5,] 0.25882546 0.14375511 -0.72595939 0.11288932 -0.55384501 -0.25659019
## [6,] 0.03538335 -0.96300206 -0.07981758 0.24464677 -0.06297391 -0.03444497
```

```
cat('Percentage of total variance contributed by each eigenvalue: \n')
```

```
## Percentage of total variance contributed by each eigenvalue:
```

```
(evals/sum(evals))*100
```

```
## [1] 68.8564344 11.5640614 10.0028749 5.6680421 3.2519810 0.6566061
```

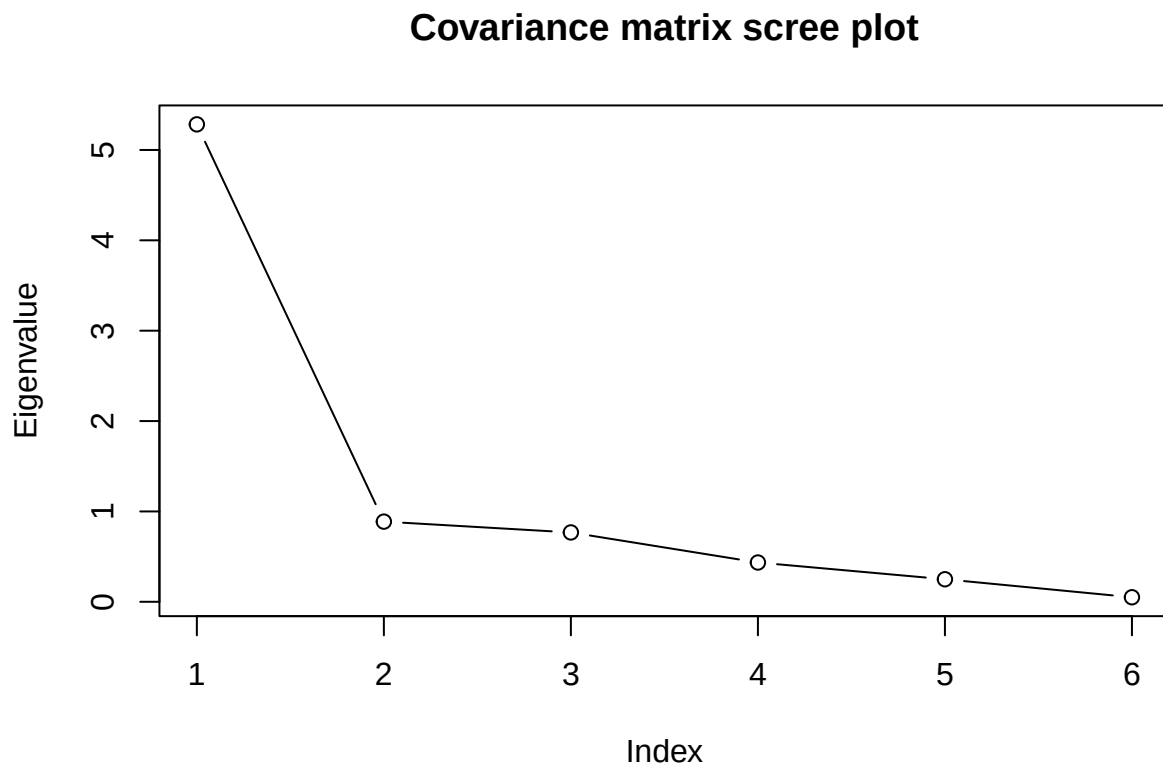
```
pca2 = princomp(S)
summary(pca2)
```

```
## Importance of components:
```

```
##               Comp.1      Comp.2      Comp.3      Comp.4      Comp.5
## Standard deviation    1.6105291 0.35119938 0.19590965 0.13271907 0.081780218
## Proportion of Variance 0.9330808 0.04437002 0.01380682 0.00633649 0.002405906
## Cumulative Proportion 0.9330808 0.97745078 0.99125760 0.99759409 1.000000000
##               Comp.6
## Standard deviation    2.184145e-09
## Proportion of Variance 1.716110e-18
## Cumulative Proportion 1.000000e+00
```

Problems 2c and 2d

```
# c
plot(seq(1:6), evals, type='b', xlab='Index', ylab='Eigenvalue',
     main='Covariance matrix scree plot')
```



```

# d
# Based on the eigenvalue variance contribution and scree plot, one might choose
# to utilize the first two sample principal components as they will explain the
# vast majority of the variance in  $x$ . It may be possible to summarize the entire
# dataset with only one principal component, as the first component was seen to
# explain 93% of the total variability. That said, doing so would omit nearly
# 7% of the variability, so a final decision depends on the level of precision
# desired. Additionally, it can be argued that three components should be used,
# despite the elbow point occurring at the second index in the scree plot. This
# is because the first two eigenvalues contribute 80% of the total variance,
# and including the third increases this to 90%. Additionally, the first three
# principal components would explain more than 99% of the variability, much more
# precise than the first two alone.

```

Problem 2e

```

corr = data.frame() # empty dataframe
for (i in 1:6) {
  for (j in 1:3) { # decided to keep three principal components
    corr[i,j] = evecs[i,j]*sqrt(evals[j]/S[i,i])
  }
}

# The rows in the table represent the variables  $x_i$ , while the columns represent
# the principal components. For example, it can be seen there is a near-exact
# direct correlation between the first variable (symptoms) and the first
# principal component (Y1), equaling 0.98919.

```